MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 Admission Onwards)

MG4DSCMAT203 Mathematical Tools for Computing

Duration: 2 hrs **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Interest (I), Appreciation (Ap), and Skill (S)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

[Learning Domain] [CO No(s)]

Part A

Short Answer Type Questions

Answer any five questions. Each question carries 2 marks

1. State the Fundamental theorem of arithmetic [U] [1]
2. Evaluate 13 mod 3 and -97 mod11 [A] [1]
3. Find the value of (1.1) + $(\overline{1.0})$. $\overline{(0+1)}$ [A] [2]
4. How many different Boolean functions of degree n are there? [K] [2]
5. Define symmetric relations. Give an example of a relation on A= {1,2,3} which is neither symmetric nor antisymmetric [U][3]
6. Is the relation R={(0,0),(0,2),(2,0),(2,2),(2,3),(3,2),(3,3)} on the set {0,1,2,3} an equivalence relation? [U][3]
7. How many edges does a tree with 100 vertices have? [K][4]
8. Draw the binary search tree (BST) for the data : 23, 50,10,56, 35, 5, 15 [An][4]

Part B

Short Essay Type Questions

Answer any five questions. Each question carries 6 marks

9. Find the GCD and LCM of of a) $2^3 .3^5 .7^2 . 11 .13$ and $2^4 . 3^3 .5. 7. 11.$

   b) 0 and 101 [A][1]

10. A parking lot has 31 visitor spaces, numbered from 0 to 30. Visitors are assigned parking spaces using the hashing function h(k) = k (mod 31), where k is the number formed from the first three digits on a visitor's license plate. Determine the parking space assigned to each car whose license plate begins with the digits: 317, 918, 007, 100, 111, and 310. [A][1]
11. Find the values of the Boolean function represented by F(x, y,z) = $x.\bar{y} + y.\bar{z}$ [A][2]
12. Find the sum-of-products expansion for the function F(x, y, z) = $(x + z).\bar{y}$. [A][2]
13. Find the matrix representation of R^2 if R is represented by the matrix $M_R = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ [U][3]

14. Define partition of a set . List the ordered pairs in the equivalence relation R produced by the partition A1 = {1, 2, 3}, A2 = {4, 5}, and A3 = {6} of S = {1, 2, 3, 4, 5, 6} [A][3]

15. Explain the pre order, in order and post order traversals in a rooted tree with an example [U][4]

16. Use Huffman coding to encode the following symbols with the frequencies listed: [A][4]

A: 0.08, B: 0.10, C: 0.12, D: 0.15, E: 0.20, F: 0.35.

What is the average number of bits used to encode a character?

Part C

Essay Type Questions

Answer any three questions. Each question carries 10 marks

17. a) Encrypt the message " **RETURN HOME**" using the Caesar cipher

a) Decrypt the message encrypted using the shift cipher f (p) = (p + 10) mod 26 [A][1]

" **DSWO PYB PEX** "

18. Design a circuit for a light fixture controlled by three switches. [An] [2]

19. Let R be the relation defined on A= {1,2 ,3,4,5,6 } by (a ,b) Ɛ R iff a ≡ b(mod 3). Then

(i) List the elements of R (ii) Show that R is an equivalence relation

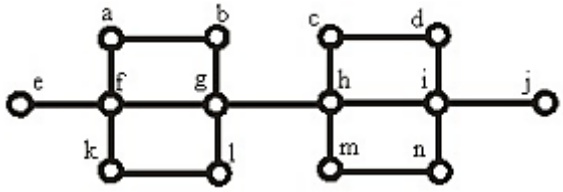
(iii) Find the equivalence class generated by the relation. [A}[3]

20. a) Define partial ordering. Show that the divisibility relation '**|**' is a partial ordering on the set of positive integers

b) Draw the Hasse diagram representing the partial ordering {(a, b)|a divides b} on A= {1, 2, 3, 4, 6, 8, 12}. [A][3]

21. a) Explain the steps of the Breadth-First Search (BFS) algorithm.

b) Find the BFS spanning tree of the following graph starting from the vertex ' a '. [A][4]



22. Explain Prim's Algorithm. Use Prim's Algorithm to find the minimal spanning tree of the following graph [A][4]

